

TECHNICAL DEEP-DIVE

Inaya Ecosystem — Developer Reference

Contract functions, API routes, data models, and SDK signatures — a lookup reference grounded directly in the real code.

Document INAYA-DEV-2026-V1 · Classification Internal · September 2026

SECTION 01

Repo & Entry-Point Map

PATH	ENTRY POINT
contracts/ (repo root)	*.sol — Hardhat project, hardhat.config.js at repo root
scripts/ (repo root)	deploy*.cjs / deploy*.js, verify-*.cjs, simulate-security-nodes.js
inaya-network-dapp/	Next.js 14 App Router — src/app/, src/lib/, src/app/api/
inaya-network-dapp/custody-sdk/	src/index.js (InayaKernel), src/crypto.js, src/contracts.js
inaya-network-dapp/custody-sdk/packages/node-daemon/	bin/inaya-node-daemon.js (commander CLI)
src/lib/chain-adapters/ (bridge + interop)	registry.js (capability model), interop/ (Wormhole)
src/lib/ai-action-requests.js, auditChain.js	AI guarded-execution state machine + cryptographic audit trail
inaya-mobile/	Expo — App.js, src/screens/, src/utils/, src/providers/
inaya-desktop/	src-tauri/src/lib.rs, tauri.conf.json
inaya-dapp-desktop/	src-tauri/src/lib.rs, tauri.conf.json

SECTION 02

Contract Function Reference

InayaStaking.sol

Constructor(stakingToken, rewardToken) — both = \$INAYA address.

- stake(uint256 amount, uint256 lockPeriodDays) — lockPeriodDays ∈ {0,30,90}, multiplier 1.00x/1.25x/1.50x, locked in on first stake
- withdraw(uint256 amount) — reverts if block.timestamp < lockExpiry
- claimReward() / exit() (withdraw all + claim)
- earned(address) / rewardPerToken() / getUserTier() → "None"/"Standard"/"Enterprise Priority"
- owner-only: setRewardRate, fundRewardPool, setEnterpriseTierThreshold, recoverForeignToken (blocked on staking/reward token)
- events: Staked, Withdrawn, RewardPaid, RewardRateUpdated, RewardPoolFunded

InayaEgressTimelockVault.sol

Constructor(inayaToken, usdtToken, stakingContract, operationalTreasury).

- executeSemiAnnualHarvest() — public, callable by anyone once currentEpochStart + 180 days has passed; sweeps full INAYA balance to stakingContract via raw transfer (not fundRewardPool)
- forwardUSDT() — public, sweeps full USDT balance to operationalTreasury
- getNextHarvestCountdown() view
- owner-only: setStakingContract, setOperationalTreasury

InayaCorporateEscrow.sol

Constructor(usdtToken) only.

- createEscrow(address corporate, address node, uint256 totalAmount) — safeTransferFrom(caller) into escrow, MONTHS=12 fixed schedule
- releaseMonthlyPayout(uint256 scheduleId) — public, callable by anyone once due
- owner-only: setMonthSeconds (testnet-only timing override), reassignNode
- events: EscrowCreated, MonthlyReleased, EscrowClosed

InayaNodeRegistry.sol

Constructor(usdtToken, verifierWallet). SCOPE NOTE in header: metrics are coordinator-verified, NOT cryptographic proof-of-storage.

- registerNode(uint256 capacityGB) — self-serve
- updateNodeMetrics(...) — verifier-only
- queueSettlement / queueSettlementsBatch — verifier-only, computes commission (30/40/50% by Entry/Mid/Enterprise tier), does not move funds
- releaseSettlement / releaseSettlementsBatch — public, callable by anyone, only after SETTLEMENT_DELAY = 36 hours
- owner-only: setVerifierWallet, setTierThresholds, withdrawExcessReserve

InayaProofRegistry.sol

Constructor(custodyAddress) — immutable IInayaCustody reference.

- registerMerkleRoot(bytes32 fileHash, bytes32 merkleRoot, uint256 chunkCount, address node) — requires msg.sender == custody.assets(fileHash).owner; node must be pre-approved via setNodeRegistered
- verifyChunkProof(...) — onlyOwner today (backend-checked); header comment documents a planned permissionless + slashing path
- events: MerkleRootRegistered, ProofVerified, NodeRegistrationChanged

InayaCustody (interface only — IInayaCustody)

Deployed, source not tracked in this repo. Interface as referenced from InayaProofRegistry:

- assets(bytes32 fileHash) view → {owner, cidAlpha, cidBeta, size, timestamp}
- batchRegisterAssets(bytes32[] fileHashes, uint256[] fileSizes, string[] shardACIDs, string[] shardBCIDs)
- event AssetRegistered

RevenueRouter (interface only)

Deployed, source not tracked in this repo. Called via inline ABI from page.js and the Stripe webhook route:

- processCorporateInvoice(uint256 usdtAmount) external

Security Layer contracts

InayaThreatRegistry (status ledger, updated only by Reporter), InayaThreatReporter (confirmThreat(bytes32 threatId, uint8 category, uint16 confidenceBps, bytes32 contributingNodesHash) — relayer-only), InayaNodeReputation (checkpointReputation(...) — relayer-only, periodic), InayaSecurityPolicy (publishPolicy(uint256 version, bytes32 policyHash, string policyURI) — owner/relayer-only).

Oracle & Automation Layer contracts (deployed to BSC Testnet)

InayaOracleRegistry (registerSource(bytes32 id, string dataType, address submitter, uint256 updateFrequency) — owner-only; isAuthorizedSubmitter(id, address) view), InayaOracleAdapter (submitData(bytes32 id, uint256 value, uint256 reportedTimestamp) — reverts on future/stale timestamp, sub-minimum-interval, or excess deviation; getLatestData(id) / isStale(id) views), InayaAutomationRegistry (registerTask(bytes32 id, address target, bytes4 selector, string condition) — owner-only; recordExecution(id, bool success, uint256 nextEligible, bytes32 txHash) — worker/owner-only, pure audit-trail write, never forwards a call to target).

SECTION 03

custody-sdk (InayaKernel) API Reference

FUNCTION	PURPOSE
<code>generateSecureSalt()</code>	16-byte random salt via <code>crypto.getRandomValues</code>
<code>deriveVaultKey({passkey, salt, iterations=100000, algo})</code>	PBKDF2 → 32-byte AES key
<code>disperseAndSlice({file, encryptionKey})</code>	Encrypt (AES-GCM-256) + base64-midpoint-split into <code>shardAlpha/shardBeta</code>
<code>anchorToLedger(...)</code>	Calls <code>InayaCustody.batchRegisterAssets()</code> with CIDs (not raw shard bytes)
<code>retrieveAndReconstruct({connection, fileHash, passkey, fetchShard})</code>	Reads <code>assets(fileHash)</code> , fetches both shards, reassembles
<code>reconstructAndDecrypt({shardAlpha, shardBeta, passkey})</code>	Concat → split salt/IV/ciphertext → re-derive key → AES-GCM decrypt
<code>encryptForPublicKey / decryptWithSecretKey / deriveEncryptionKeyPairFromSignature</code>	X25519 + HKDF-SHA256 + XChaCha20-Poly1305 sealed-box — re-wraps the passkey for sharing, not the file
<code>connectWallet, approveFeeTokens</code>	ethers v6 wallet/allowance helpers
<code>Staking.{stake,unstake,claimReward,calculateReward,getStakedBalance}</code>	Thin wrapper over <code>InayaStaking</code>
Payments, Metadata, Analytics, events (<code>EventEmitter</code>)	Supporting modules
<code>errors.</code> <code>{InayaError,InayaValidationError,InayaWalletError,InayaContractError,InayaNetworkError}</code>	Typed error classes

Crypto primitives are `@noble/hashes`, `@noble/ciphers`, `@noble/curves` (pure JS) rather than `crypto.subtle` — chosen specifically for React Native compatibility, per an inline comment in `crypto.js`.

Release verification (v1.0.10-beta+)

Every git tag matching `v*` on `custody-sdk` (`.github/workflows/release.yml`) runs `npm test` → computes checksums → pins to IPFS (Pinata, non-blocking) → `npm publish --provenance` → commits `CHECKSUMS.md` → attaches the tarball to a GitHub Release. Reproduce independently per `docs/VERIFYING_RELEASES.md`:

- `git rev-parse <tag> ^{tree}` — compare to `CHECKSUMS.md`'s `git-tree-hash`. NOT `git archive | sha256sum` (tried first, found non-reproducible across git versions — see Section 05 of `ecosystem-architecture.js`).
- Download the .tgz attached to the GitHub Release, `sha256sum` it, compare to `CHECKSUMS.md`'s `npm-tarball-sha256` — always matches, since you're hashing the literal file CI attached rather than re-deriving it (npm pack tarball bytes aren't guaranteed identical across npm versions, even when file contents are).
- `npm view @inaya-network/custody-sdk@<version> dist.integrity / dist.shasum` — the registry's own recorded hash for the published tarball.
- `test/webCryptoCompat.test.mjs` — the committed proof that this package's crypto and the dApp's former inline `crypto.subtle` implementation are byte-identical and cross-decryptable.

SECTION 04

node-daemon CLI Reference

COMMAND	EFFECT
login	Reads INAYA_PRIVATE_KEY (env or prompt), encrypts at rest (PBKDF2-SHA256 200k iter + AES-GCM-256), writes ~/.inaya/node-daemon/config.json
register <capacityGB>	InayaNodeRegistry.registerNode(capacityGB) on-chain + POST /api/nodes/register off-chain
start	Foreground 5-min heartbeat loop → POST /api/nodes/heartbeat with {nodeId, operatorWallet, totalCapacityGB, usedCapacityGB:0, shardsStored:0} (last two hardcoded 0 — no shard-storage capability exists)
report <indicator> - -category <cat>	Security Layer signed observation → POST /api/security/report
service install / uninstall	Windows background service via node-windows

package.json description states it plainly: registers your node on-chain and reports heartbeat/telemetry. Does not store or serve shards. NODE_REGISTRY_ABI in this package has exactly two functions (registerNode, nodes) — no stake/settlement/withdraw calls exist in the daemon's on-chain surface.

SECTION 05

Backend lib/ Reference

Every file below follows the same shape: getXCollections(), ensureXIndexes() (module-level indexesEnsured guard), validateXInput() (throws, fail-closed).

FILE	DOMAIN
security.js	Threat reports/threats/policy/events/reputation-cache, computeThreatConfidence, getPublicSecurityStats
ai-security-tools.js	Security Assistant tool declarations + dispatcher
learn.js / learnConfig.js / youtube.js	Learn saved/progress/search-cache/video-cache, category+collection config, cached YouTube Data API client
ai-learn-tools.js	Learn Tutor tool declarations + dispatcher
orgs.js / orgPlans.js	Business Workspace auth (sessions, magic-links), org/plan definitions; canManageOrg/canAccessDepartment plus the additive canManageFinance/canAccessFinance/canManageHR/canAccessHR/isDepartmentManager/isSelfEmployeeRecord gates
document-workflow.js / document-permissions.js	Document state machine + activity log; access resolution + share tokens + getAccessibleScope() (the one org-wide 'what can this member see' resolver, extended per module as each shipped)
ai-business-tools.js	Business Assistant tool declarations + dispatcher (permission-aware) — list_documents/tasks/contacts/deals/suppliers/purchase_orders/products/invoices/expenses/employees, find_employee_document
task-workflow.js / deal-workflow.js / purchase-request-workflow.js / purchase-order-workflow.js	Business Operations transition-table workflows — {from,to,requiresManage,activityAction} pattern, one findOneAndUpdate per transition
inventory.js	recordStockMovement(), getStockLevel(), totalStockForProduct(), isLowStock() — stock_levels is \$inc-only, stock_movements is the append-only ledger
invoice-workflow.js / expense-workflow.js	transitionInvoice() (send/markPaid/cancel + cron-only markOverdueInvoices()), transitionExpense() (submit/approve/reject/cancel)
employee-workflow.js / leave-workflow.js	transitionEmployee() (onboarding→active→on_leave→terminated), transitionLeaveRequest(), getLeaveBalance() (computed fresh, never a stored counter)
attachments.js	createAttachment()/listAttachmentsForRecord()/serializeAttachment() — shared metadata layer for Finance receipts (EXPENSE) and HR documents (EMPLOYEE), same encrypt/shard/pin client pipeline as org_documents
org-activity-log.js	logOrgActivity()/listOrgActivityForRecord() — one domain-generic audit log every Business Operations/Finance/HR workflow writes into
rag/collections.js, chunking.js, embeddings.js, sanitize.js, vectorStore.js, retrieve.js, ingest.js, youtubeTranscript.js, queryCache.js, metrics.js	RAG pipeline — see Section 11 for function-level detail
rag/sources/{docsSources,securitySources,learnSources}.js	Per-domain source adapters feeding ingest.js — DOCS_SOURCES, SECURITY_SOURCES, LEARN_STATIC_SOURCES + ensureVideoTranscriptIngested()
dataroom.js	Investor Data Room visitors/sessions/documents/views
activity.js	DAU/WAU ping recording + stats
watcherPioneer.js	Watcher Pioneer enrollment/qualification
feedback.js	Bug reports / idea submissions
metadata-auth.js	Shared signed-message verification helper (verifyMetadataAuth)
email.js	Resend-backed transactional email (magic links, Data Room verify)

SECTION 06

API Route Reference (by domain)

DOMAIN	ROUTES
Security (public)	GET /api/security/{threat,feed,stats,policy}, POST /api/security/report, GET /api/security/reputation/[address], POST /api/security/events, GET /api/security/events
Security (admin)	GET /api/admin/security/{threats,nodes}, POST /api/admin/security/threats/[id]/override, GET POST /api/admin/security/policy
Security (AI)	POST /api/ai/security-chat
Security (cron)	GET /api/security/cron/checkpoint-reputation (CRON_SECRET-gated)
Learn	GET /api/learn/config, GET /api/learn/search, GET /api/learn/video/[videoid], GET POST /api/learn/saved, DELETE /api/learn/saved/[videoid], GET POST /api/learn/progress, POST /api/learn/report, POST /api/learn/analytics
Learn (AI)	POST /api/ai/learn-chat
Business Workspace (auth)	POST /api/orgs/create, /api/orgs/login/google, magic-link login/verify
Business Workspace (org)	department/project CRUD, document upload/list/retrieve/transition, permissions grant/update/revoke, share create/list/revoke, GET /api/orgs/share/[token] (public resolve)
Business Workspace (billing)	GET /api/orgs/billing, POST /api/orgs/billing/checkout, POST /api/orgs/billing/portal, GET /api/orgs/billing/plans
Business Workspace (AI)	POST /api/ai/business-chat
Business Workspace (tasks/CRM/procurement/inventory)	/api/orgs/tasks/**, /api/orgs/crm/{contacts,deals}/**, /api/orgs/procurement/{suppliers,requests,orders}/**, /api/orgs/inventory/{products,warehouses,movements}/**, GET /api/orgs/dashboard (aggregate summaries per module)
Business Workspace (finance)	/api/orgs/finance/invoices/** (incl. [id]/transition, [id]/activity), /api/orgs/finance/expenses/** (incl. [id]/attachments, [id]/transition), /api/orgs/finance/payments/** (incl. [id]/approve), GET /api/orgs/finance/reports (?type=revenue expenses outstanding paid-unpaid&format=json csv)
Business Workspace (HR)	/api/orgs/hr/employees/** (incl. [id]/attachments, [id]/leave-balance, [id]/transition, [id]/activity), /api/orgs/hr/leave-requests/** (incl. [id]/transition), POST DELETE /api/orgs/hr/departments/[id]/manager
Business Workspace (finance/HR cron)	GET /api/cron/invoices-mark-overdue (CRON_SECRET-gated, nightly SENT→OVERDUE flip)
RAG (public, via existing assistants)	Consumed inside POST /api/ai/chat (Docs), /api/ai/security-chat, /api/ai/learn-chat — no separate public RAG endpoint
RAG (admin)	GET /api/admin/rag/stats, POST /api/admin/rag/reingest
RAG (cron)	GET /api/cron/rag-reingest (CRON_SECRET-gated)
Data Room (public)	POST /api/dataroom/request-access, GET /api/dataroom/verify, GET /api/dataroom/session, POST /api/dataroom/accept-nda, GET /api/dataroom/documents, GET /api/dataroom/documents/[id]/stream, POST /api/dataroom/documents/[id]/view-event
Data Room (admin)	GET POST /api/admin/dataroom/documents, DELETE /api/admin/dataroom/documents/[id], GET /api/admin/dataroom/visitors, POST /api/admin/dataroom/visitors/[id]/revoke

DOMAIN	ROUTES
Nodes	POST /api/nodes/register, POST /api/nodes/heartbeat, GET /api/nodes/settlements/release (cron)
Referrals	POST /api/referrals/{activate,initiate,webhook}, GET /api/referrals/{status,history,leaderboard}
Watcher Pioneer	POST /api/watcher/{enroll,qualify}, GET /api/watcher/status
Activity	POST /api/activity/ping, GET /api/admin/activity
Other	POST /api/faucet, POST /api/stripe-webhook, POST /api/create-payg-checkout-session, POST /api/feedback/submit, POST /api/feedback/upload, GET /api/admin/feedback

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SECTION 07

MongoDB Collections Reference

COLLECTION GROUP	KEY COLLECTIONS
Security	security_reports, security_threats, security_policy, security_events, security_reputation_cache
Learn	learn_saved, learn_progress, learn_search_cache (TTL 24h), learn_video_cache (TTL 24h), learn_reports, learn_analytics_events
Business Workspace (core)	orgs, org_members, sessions, magic_links, departments, projects, documents, document_permissions, document_shares, document_activity, project_members
Business Operations	tasks, crm_contacts, crm_deals, suppliers, purchase_requests, purchase_orders, warehouses, products, stock_levels, stock_movements, org_activity (one shared domain-generic audit log for every module below, not just Business Operations)
Finance & HR	invoices, expenses, payments, employees, leave_requests, attachments (shared metadata store — EXPENSE and EMPLOYEE relatedRecordType)
RAG	rag_chunks (with Atlas Vector Search + Atlas Search indexes), rag_sources, rag_ingestion_runs, rag_query_log, rag_embedding_cache
Data Room	dataroom_documents, dataroom_visitors, dataroom_magic_links, dataroom_sessions, dataroom_views
Activity	activity_pings — one doc per {surface, identityId, date}, unique compound index
Referrals / Watcher Pioneer / Feedback	referrals, watcher_enrollments, feedback_submissions (each with their own lib.js as listed in Section 05)

SECTION 08

AI Assistant Tool-Calling Pattern

Identical shape across all four assistants (src/app/api/ai/{chat,business-chat,security-chat,learn-chat}/route.js)
— chat/route.js is the Docs Assistant, the other three are named for their domain:

- Model: gemini-flash-latest, thinkingConfig: { thinkingLevel: "low" }, maxOutputTokens: 800
- MAX_TOOL_ROUNDS = 5 — a bounded loop of generateContent → check functionCalls → run tools → feed functionResponse back in
- generateContentWithRetry() — retries on HTTP 429/503 with a [700ms, 1800ms] backoff schedule
- Route-level: maxDuration = 90, not streaming (deliberate — unpredictable number of tool round-trips before a final answer)
- Per-assistant: buildXContext() computed once per request, X_TOOL_DECLARATIONS (Gemini Type.OBJECT schemas), runXTool(name, args, ctx) dispatcher, xSystemInstruction() builder

SECTION 09

Auth Patterns — Technical Detail

PATTERN	IMPLEMENTATION DETAIL
Wallet signed-message	ethers.verifyMessage(reconstructedMessage, signature) === expectedAddress, 5-minute timestamp freshness window, reimplemented per-feature (metadata-auth.js has the canonical version, security.js/watcherPioneer.js reimplement locally by convention)
Magic-link	generateToken() (random), hashToken() (stored hashed, never plaintext), TTL-indexed Mongo collection, consumeLoginToken() atomic findOneAndUpdate to prevent replay
Session (web)	HttpOnly cookie, createSession(email) helper in orgs.js
Session (mobile)	Bearer token, same session concept, stored via expo-secure-store
Share tokens	generateShareToken()/hashShareToken(), atomic findOneAndUpdate-based single-consume pattern (document-permissions.js), reused as a template for Data Room's magic-link tokens

SECTION 10

Testing & Verification Conventions

- Hardhat tests: `test/security-layer.test.js` (17 tests — access control, confirm/re-confirm, false-positive override, 4-signer simulation)
- Node backend tests: `test/*.test.mjs` run via `node --test` against a real MongoDB instance (not mocked) — `security.test.mjs` (11), `learn.test.mjs`, `dataroom.test.mjs`, `activity.test.mjs`, `feedback.test.mjs`, `watcher-pioneer.test.mjs` — 140/140 passing as of the Security Layer build, no regressions
- Business Operations: `business-operations-tasks.test.mjs`, `crm-workflow.test.mjs`, `procurement-workflow.test.mjs` (incl. real inventory-ledger integration — a receipt genuinely moves stock), `document-permissions.test.mjs`
- Finance & HR: `finance-workflow.test.mjs` (8 tests — invoice lifecycle, cron overdue-flip + idempotency, org isolation, expense approval gate, payment record→approve replay-safety), `hr-workflow.test.mjs` (9 tests — employee lifecycle, leave approval + computed-balance correctness, self-access vs. HR-role vs. Department Manager scope) — both caught real bugs before shipping: a race condition in PO receiving (stock movements were applying before the PO's own optimistic-concurrency-guarded update), and a Department Manager visibility bug in `getAccessibleScope()` (Section 12)
- RAG: `rag-ingestion.test.mjs`, `rag-security.test.mjs`, `rag-attribution.test.mjs`, `rag-learn-transcript.test.mjs`, `rag-retrieval.test.mjs` (28 tests total) — run against the real Atlas cluster and real Gemini embeddings, not mocked; the relevance threshold (Section 11) was empirically calibrated from live measurements taken during this test work, not guessed
- On-chain simulation: `scripts/simulate-security-nodes.js` — 4 throwaway signers report a clearly-fake test indicator end-to-end through to a real on-chain `confirmThreat()` transaction on BSC Testnet
- Manual browser verification is the norm for UI work in this codebase — `curl/fetch` checks for API correctness, then a real browser pass (or explicit statement that one hasn't happened yet) before anything is called "verified"

SECTION 11

RAG Pipeline — Function Reference

FUNCTION	FILE	PURPOSE
chunkMarkdownByHeading() / chunkParagraphs() / chunkStructuredSections() / chunkQaPairs()	rag/chunking.js	Source-shape-specific chunkers — structured-sections for fundraising-docs' {sections: [[blocks]]} shape, qa-pairs for FAQ arrays, heading-split for markdown, paragraph-split as the fallback
embedText() / embedChunkText() / embedQueryText()	rag/embeddings.js	Wraps Gemini's models.embedContent (gemini-embedding-001, 768 dims) — RETRIEVAL_DOCUMENT taskType for ingestion, RETRIEVAL_QUERY for search; content-hash-cached in rag_embedding_cache
sanitizeChunkText() / wrapContextBlock()	rag/sanitize.js	Injection-pattern neutralization before a chunk is ever fed into a model prompt, and delimited wrapping so retrieved context can't be confused with the system prompt
ensureIndexes() / upsertChunks() / deleteBySource() / deleteObsoleteChunks() / hybridSearch()	rag/vectorStore.js	Idempotent index creation (\$vectorSearch + \$search via collection.createSearchIndex()); hybridSearch() runs both queries and merges via Reciprocal Rank Fusion in application code
retrieveContext() / formatAttribution()	rag/retrieve.js	The one function every assistant calls. DEFAULT_MIN_RELEVANCE = 0.8 — hasResults gates purely on top vector score, no keyword-match override (removed after it let one irrelevant query pass)
ingestSource() / ingestAllStaticSources() / deleteSource()	rag/ingest.js	Content-hash diffing per source: insert new chunks, update changed ones, skip unchanged ones, remove chunks whose source content no longer produces them
fetchYouTubeTranscript(videoid)	rag/youtubeTranscript.js	Scrapes the public timedtext caption-track URL out of a video's watch-page HTML — unofficial (the official Data API's captions.download needs OAuth as the channel owner), always fails gracefully to null, never throws
recordRetrieval() / recordIngestionRun()	rag/metrics.js	Writes to rag_query_log / rag_ingestion_runs — fail-open, a metrics write never blocks the actual retrieval/ingestion it's recording

DOCS_SOURCES (rag/sources/docsSources.js) covers inaya-knowledge.js, business-workspace-guide.md, the FAQ page's faqs export, and all 15 scripts/fundraising-docs/content/*.js files — explicitly excluding custody-sdk's own docs (a nested, separately-git-excluded repo — see Section 20's build-breakage history for why touching it at all is risky) and the legacy KNOWLEDGE_ARTICLES collection.

SECTION 12

Business Operations & Finance/HR — Workflow Reference

Every transition below follows the same shape: one `findOneAndUpdate({..., status: fromState}, {$set: {status: toState}})` — a mismatched current status is a clean 409, no extra locking needed.

MODULE	TRANSITIONS
task-workflow.js	submit: DRAFT→IN_PROGRESS, complete: IN_PROGRESS→DONE, reopen: DONE→IN_PROGRESS, cancel: *→CANCELLED
deal-workflow.js	stage transitions across the pipeline, terminal: →WON / →LOST
purchase-request-workflow.js	submit: DRAFT→PENDING_APPROVAL, approve/reject: PENDING_APPROVAL→APPROVED REJECTED (requiresManage), cancel: DRAFT PENDING_APPROVAL→CANCELLED
purchase-order-workflow.js	submit→approve→order→receive (partial and full) — <code>receivePurchaseOrder()</code> applies inventory stock movements only AFTER the PO's own optimistic-concurrency-guarded status update succeeds (reordered after a real race condition was caught — see Section 10)
invoice-workflow.js	send: DRAFT→SENT, markPaid: SENT OVERDUE→PAID, cancel: DRAFT SENT OVERDUE→CANCELLED, markOverdueInvoices() (cron-only): SENT→OVERDUE
expense-workflow.js	submit: DRAFT→PENDING_APPROVAL, approve/reject: PENDING_APPROVAL→APPROVED REJECTED (requiresManage: <code>canManageFinance</code>), cancel: DRAFT PENDING_APPROVAL→CANCELLED
employee-workflow.js	activate: ONBOARDING→ACTIVE, placeOnLeave: ACTIVE→ON_LEAVE, returnFromLeave: ON_LEAVE→ACTIVE, terminate: ACTIVE ON_LEAVE→TERMINATED (requiresManage: <code>canManageHR</code>)
leave-workflow.js	approve/reject: PENDING→APPROVED REJECTED (requiresManage: <code>canManageHR</code>), cancel: PENDING→CANCELLED (requester or <code>canManageHR</code>); <code>getLeaveBalance()</code> = <code>annualLeaveAllocationDays</code> – <code>sum(this year's APPROVED day-spans)</code> , computed on every call

getAccessibleScope() extension pattern. Each module adds its own `visibleX` array (`visibleTasks`, `visibleContacts`, `visibleInvoices`, `visibleEmployees`, ...) to the same resolver rather than each having a separate scope function. Finance's arrays are gated by `canAccessFinance` on top of department scope; HR's `visibleEmployees` unions HR-scoped employees, Department-Manager-scoped employees (`managedDepartmentIds` — NOT filtered against the caller's own already-visible departments, a real bug fixed during this layer's testing, see the founder architecture doc's Section 24), and the caller's own self-record, deduplicated by `_id`.

SECTION 13

Multi-Chain Bridge & Interop Reference (September 2026)

FUNCTION / CONTRACT	PURPOSE
InayaTokenBridgeHome.bridgeOut(...)	BSC-side lock/burn entry point, source of every cross-chain transfer
InayaTokenBridgeSpoke.executeMessage(...) / receive_message(...)	Remote-chain entry point (EVM name / Solana-Move name) — validates threshold validator signatures, mints/releases on the destination
InayaChainRegistry.setSpokeBridgeAddress() / registerRemoteChain() / setTrustedRemoteContract()	Owner-only wiring — a chain has no real route until all three are called for it, not just deployed
listChainCapabilities() / getChainCapability(chainId)	src/lib/chain-adapters/registry.js — the single source a UI reads before ever describing a chain as 'supported'; SUPPORT_LEVELS run DISCOVERED(0) → READ_ONLY → WALLET → MESSAGE → TOKEN_TRANSFER → STAKING → FULL_ECOSYSTEM(6), each backfilled only from what's actually been verified
WormholeProvider.js	Wraps Wormhole's WTT/lock-and-mint flow; chain-id mapping is the thing that broke first (Sepolia/testnet IDs differ from their mainnet counterparts unlike BSC/Avalanche)
GET/POST /api/interop/wtt/{initiate,relay,status}	Relay cron (* / 5 min) fetches the Guardian-signed VAA via a plain REST call to Wormholescan's public API (not the @wormhole-foundation/sdk package — that broke the Next.js build, see Section 26 of the architecture doc) and completes the transfer via Inaya's relayer wallet

Per-chain proof records (real tx hashes both sides):

deployments/bridge/{avalancheFuji,solanaDevnet,hederaTestnet,aptosTestnet,suiTestnet}.json;

deployments/interop/wormhole-wtt/bscTestnet-attestation.json.

SECTION 14

Storage Backup & Redundancy Reference (September 2026)

FUNCTION / CONTRACT	PURPOSE
InayaBackupRegistry.sol	0x062c341aE4f11CB1dEa1B0D3930d52902F97f48a (BSC Testnet) — records redundancy commitment + health-state transitions only, never replica data
InayaKernel.Backup.getBackupStatus / getBackupHealth / getRedundancyStatus / getRecoveryStatus / requestRecovery	custody-sdk client, published to npm
Health states	PROTECTED / REBUILDING / DEGRADED / RECOVERY_REQUIRED / RECOVERY_FAILED — pure state machine, 19/19 unit tests
Pin-status check	Every 15 min, 3-strike grace window before flagging a replica
Integrity check	Daily, content-hash comparison, zero grace — catches real corruption immediately

Providers: Pinata + Filebase (S3-compatible, IPFS-backed, separate infrastructure/failure domain). Filebase gotcha: its S3 endpoint is s3.filebase.io, not .com; the AWS SDK's default auto-attached request checksum causes a generic AccessDenied against it unless disabled. Full writeup: docs/backup-redundancy-architecture.md.

SECTION 15

AI Guarded Execution & Audit Trail Reference

(September 2026)

FUNCTION	PURPOSE
proposeAiAction({orgId, assistantSurface, toolName, targetRecordType, targetRecordId, proposedAction, args, actorEmail, canPropose})	Inserts a PENDING_APPROVAL request; idempotencyKey = sha256(orgId+toolName+targetRecordId+args+hourBucket) dedupes same-hour repeats
reviewAiAction(..., canApprove)	Approve sets unlockAt = now + 36h (SETTLEMENT_DELAY_MS); reject is terminal
classifyRisk(targetRecordType, proposedAction)	Never returns undefined — unmatched combinations default MEDIUM; EXPENSE/INVOICE/PURCHASE_ORDER default HIGH
api/cron/execute-approved-ai-actions	The only code path that calls a real transitionX() for an AI-originated change, and only once unlockAt has passed; runs as a synthetic org-manager membership, attributed in the log to the human approver
appendAuditEntry({orgId, recordType, recordId, actorEmail, action, previousState, newState, metadata})	src/lib/auditChain.js — entryHash = sha256(prevHash + canonicalJSON(eventFields)); optimistic CAS on audit_chain_heads, retries on conflict, never corrupts under contention
verifyChainIntegrity(orgId)	Walks the full chain, recomputes every hash; returns {valid:false, brokenAtSeq, reason} on the first mismatch (gap/reorder, prevHash mismatch, or content altered after write)
GET /api/orgs/audit/export?orgId=&format=json csv	Org-scoped (owner/admin only) full-chain export — every field needed to independently recompute and verify, so a customer never has to trust a UI badge

9 gated domains (EXECUTORS map in ai-action-requests.js): TASK, EXPENSE, DOCUMENT, EMPLOYEE, INVOICE, LEAVE_REQUEST, PURCHASE_ORDER, PURCHASE_REQUEST, DEAL — one corresponding propose_* tool each in ai-business-tools.js. State machine: PENDING_APPROVAL → APPROVED → QUEUED → EXECUTED|EXPIRED, or PENDING_APPROVAL → REJECTED, or APPROVED → CANCELLED (only before unlockAt). 19 automated tests, 11 adversarial.

SECTION 16

Web3 App Store, NFT Vault & MFA Reference
(September 2026)

FUNCTION / ROUTE	PURPOSE
submitAppListing / listApprovedListings / listPendingListings / listListingsBySubmitter / reviewAppListing	src/lib/appStoreListings.js — full submission→threat-check→review→approve pipeline
POST /api/apps/submit, GET /api/apps/listings, GET /api/apps/my-listings	Developer-facing App Store API
GET /api/admin/app-store/pending, POST /api/admin/app-store/[slug]/review	Admin review queue, passphrase-gated same as /admin/audit
src/app/apps/embed/[slug]/page.js	Sandboxed iframe host for externally-hosted app submissions — no allow-same-origin + allow-scripts together
checkNftContractSupport / discoverOwnedTokens	src/lib/nftDiscovery.js — ERC-721 + Enumerable only, no indexer
signNftBackup	src/lib/nftBackupAuth.js — wallet-signature auth reused for NFT backup requests
POST /api/nft/backup, GET /api/nft/backups	NFT Vault backup API
src/components/AddressRiskCheck.js	Debounced live Security Layer threat-registry check, wired into the Bridge page's recipient field
MfaSettings.js / MfaVerifyScreen.js	TOTP + Firebase-Phone-Auth SMS MFA for Business Workspace login

SECTION 17

Sovereign Enterprise OS Reference (September 2026)

MODULE	KEY EXPORTS / FUNCTIONS
src/contexts/OrgContext.js	OrgProvider, useOrg() → {email, membership, orgId, role, orgs, switchOrg, logout, can: {manageOrg, accessDepartment, manageFinance, accessFinance, manageHR, accessHR, isDepartmentManager}}
src/contexts/WalletContext.js	WalletProvider, useWallet() → {walletAddress, walletBalance, isConnected, selectedWalletName, signMessage}
src/lib/orgGates.js	Client-safe extraction of orgs.js's 7 pure permission gates — re-exported from orgs.js under the same names for the ~60 existing server-side importers
src/lib/trustHealth.js	computeTrustHealthSnapshot(({scope:"org" "wallet", ...}) → {overallStatus, auditTrail security, aiActions backup, businessHealth, scopeNotes}
GET /api/orgs/trust-health, GET /api/wallet/trust-health	Org: requireMembership-gated. Wallet: unauthenticated, aggregate counts only
src/lib/notifications.js	createNotification (idempotent on dedupeKey), listNotificationsFor, markRead, markAllRead — collections: notifications, notification_reads
GET/POST /api/orgs/notifications/*, /api/wallet/notifications/*	Org: session-cookie auth. Wallet: GET unauthenticated, mutations signed via verifyMetadataAuth's existing message format
src/lib/orgSearch.js, src/lib/walletSearch.js	searchOrg(({orgId, membership, email, query}) over getAccessibleScope(); searchWallet(({walletAddress, query}) over metadata_files
src/components/CommandPalette.js	Cmd/Ctrl+K overlay, surface-agnostic — onSelect(result) left to the caller's own real navigation function
src/lib/activityCenter.js	generateWhatChanged(({scope, ..., period}) → {sections: [{module, bullets}]}); reuses business-brief.js's BRIEF_PERIODS
src/lib/ai-os-router.js	buildOsContext, getOsToolDeclarations(scope), runOsTool(name, args, ctx), osSystemInstruction — business_/security_ prefixed dispatch
POST /api/ai/os-chat, POST /api/ai/os-chat-wallet	Org: requireMembership-gated. Wallet: identityId passed directly, matching security-chat's existing precedent
src/components/business/OsHomeView.js, src/components/OsHomeSection.js	The composed OS Home screen per surface — TrustHealthCard, OS Assistant widget, What-Changed preview, module tiles, surfaced links
open_module_window(label, path)	Tauri command, both inaya-desktop and inaya-dapp-desktop's src-tauri/src/lib.rs — pops a module into its own native window

Every org-scoped route follows the exact ensureOrgIndexes()+requireMembership()+getAccessibleScope()-or-narrower pattern dashboard/route.js already established; every wallet-scoped GET follows the same unauthenticated-aggregate-only trust tier as the pre-existing list-files/backup-status routes. No existing route, collection, or component was renamed, removed, or had its behavior changed — additive only.

SECTION 18

Healthcare & Legal OS Reference (September 2026)

FUNCTION / ROUTE	PURPOSE
health-patients.js: createPatient, assignCareTeamMember, findDuplicatePatientCandidates, mergePatients	Patient registry; care-team assignment is the join row getAccessibleScope() reads for visibility
health-clinical-workflow.js: TRANSITIONS (draft→review→sign→lock), amendClinicalRecord	Locked records are never overwritten — amend creates a new linked record, marks the original AMENDED
health-consent-workflow.js, health-roi-workflow.js, health-breakglass.js	Consent record/withdraw; ROI request→authorize→review (hands off to export-center.js); break-glass = a real time-limited care-team assignment, not a bypass path
legal-matter-workflow.js: createMatter, assignMatterTeamMember; legal-conflict-workflow.js: searchConflicts, recordConflictCheck	Matter lifecycle + team assignment; conflict status is always potential/cleared/escalated, never a raw boolean
legal-evidence.js, legal-custody.js: recordCustodyEvent; legal-hold-workflow.js: createLegalHold, releaseLegalHold	Evidence metadata over an org_documents FK; every custody event chained via appendAuditEntry; holds gate retention.js's checkDispositionAllowed()
trust-accounting.js: recordDeposit, recordWithdrawal, getMatterTrustBalance	Atomic balance recheck before every withdrawal — no jurisdictional (IOLTA) compliance claimed
ai-health-tools.js, ai-legal-tools.js	buildXContext/X_TOOL_DECLARATIONS/runXTool/xSystemInstruction — read/summarize/draft only, wired into ai-os-router.js with health_/legal_ prefixes
industry-config.js: requireVertical(orgId, expectedVertical)	The door-lock — every Health/Legal route calls this right after requireMembership(); 403 on any mismatch, fail-closed as 'general' with no vertical set
25 routes under /api/orgs/health/*, /api/orgs/legal/*	Patients, consents, ROI, break-glass, appointments, billing, care-team, research (health); clients, prospects, matters, matter-team, conflict-checks, evidence, holds, discovery, redaction, deadlines, contracts, entities, trust-accounting, time-entries, billing (legal)
HealthView.js, LegalView.js (src/components/business/)	The only two modules with an actual screen today — registry/list + create + detail modal, mirroring CRMView.js's structure
src/lib/integrations/{health,legal}/*Adapter.js + adapterStub.js	13 documented third-party interfaces (FHIR, HL7, DICOM, lab, pharmacy, claims, e-filing, legal research, e-signature, email, calendar, accounting, SSO) — configured:false until real env credentials are present, never fabricated

Patient/matter visibility is assignment-based (health_care_team_assignments/legal_matter_team_assignments), a genuinely different model from every other Business Workspace module's department-scoping — wired directly into getAccessibleScope() rather than a parallel resolver. 20 domain modules have real backend + API; only 2 (Patients, Matters) have a screen. Two real bugs were caught and fixed during this build (break-glass had no expiry enforcement; 14 POST route handlers referenced an out-of-scope variable after the vertical-lock was added) — both now covered by permanent regression tests.

SECTION 19

Financial Services & Regulated Enterprise OS Reference (2026)

FUNCTION / ROUTE	PURPOSE
compliance-controls.js, compliance-evidence.js, control-testing.js	Control library + evidence vault; a failed control test auto-opens a Finding via the shared finding lifecycle
compliance-policies.js: createPolicyDraft→submitForReview→approve→publishPolicy, amendPolicy	Publish sets immutable:true; amendPolicy is the ONLY path forward from PUBLISHED, always inserting version+1, never mutating the published row
regulatory-examination-access.js: createExaminerMagicLink, exchangeMagicLink	Scoped (examinationId, requestId) access, deliberately its own token pattern (not a retrofit of the unrelated investor dataroom.js)
financial-entities.js, financial-funds.js: isFundTeamMember	Entity-scoped fund visibility — same assignment-based shape as isCareTeamMember/isMatterTeamMember
investment-committee.js, cap-table.js, board-management.js	IC decision lifecycle (finalized decisions are versioned, never overwritten); cap-table is an ingest/evidence layer, never a transactional share registry
privileged-access.js: requestElevation→approveElevation, grantBreakGlass, reviewSession	Cross-vertical from day one — the approver must be a different person than the requester; break-glass is ACTIVE immediately but always forces a post-event review
integrations.js: INTEGRATION_PROVIDERS, configureIntegration, recordSyncRun	A connection can only ever reach ACTIVE through a real recorded sync run — never through configuration alone
ai-audit-tools.js, ai-regulatory-tools.js + 7 more role-specific copilots	All 100% read/summarize/flag — a compliance-certification request is refused at the tool layer before any summarization runs
~20 API subtrees under /api/orgs/{regulated,financial,private-capital}/*	Every route: ensureOrgIndexes → requireMembership → requireVertical → lib call, statically verified by vertical-lock-wiring.test.mjs
RegulatedView.js, FinancialView.js (src/components/business/)	Full tab-router UI — Controls, Evidence, Findings, Policies, Risk, Audits, Examinations, Dashboard (Regulated); equivalent depth for Financial

All ten SOW phases are live-verified end-to-end on web: control creation/testing/finding lifecycle, the policy publish-immutability guard, an examiner magic-link's one-time-use cycle, entity-scoped fund/deal visibility, and the board-report/export-package immutability guards. Nothing on inaya-mobile for any of these three verticals yet. Every third-party integration (fund admin, custodian, KYC/AML, etc.) is a documented, honest not-configured stub.

SECTION 20

Government & Public Sector Sovereign OS Reference (2026)

FUNCTION / ROUTE	PURPOSE
citizen-records.js: createCitizenRecord, assignCitizenRecord, requireCitizenRecordAccess	requireCitizenRecordAccess() is the load-bearing gate — department membership or governmentRole:'staff' alone is never enough, only an explicit assignment (or org owner/admin) grants access to one specific record
government-cases.js: CASE_STATES (OPEN→ASSIGNED→IN_PROGRESS→PENDING_REVIEW→RESOLVED→CLOSED)	Same TRANSITIONS-map + atomic findOneAndUpdate template as incidents.js; a case linked to a citizen record inherits that record's need-to-know check
policy-knowledge-base.js: createEntryDraft→publishEntry, amendEntry	compliance-policies.js's exact publish-immutable lifecycle, recontextualized — no updateEntryDraft() path reachable once PUBLISHED
government-audit.js: logCitizenRecordAccess, logGovernmentDocumentRead	Government-vertical-only: every document READ is chain-logged, not just writes — an additive branch inside the existing retrieve route, gated so no other vertical's behavior changes
government-dashboard.js: getGovernmentDashboard	Operations + security readiness as two separate honest panels — a fresh org with zero audit-chain entries reports 'unknown,' never a fabricated 'valid'
ai-government-tools.js: search_citizen_records, summarize_case, get_dashboard_summary	100% read-only; need-to-know is re-checked INSIDE the tool per record, not just trusted from the API layer above it
privileged-access.js (reused, zero new code)	Break-glass access for Government orgs — already cross-vertical since §33, no government-specific fork needed
integrations.js: 5 new government_systems providers	Civil/national ID registry, legacy government ERP, GIS/land records, public records portal, interagency data exchange — all configured:false stubs
11 routes under /api/orgs/government/*	citizen-records (+assign), cases (+transition), policy-kb (+transition/publish/amend), dashboard — statically verified by the extended vertical-lock-wiring.test.mjs

FUNCTION / ROUTE	PURPOSE
GovernmentView.js (src/components/business/)	Tab-router UI — Citizen Records, Cases, Policy Knowledge Base, Dashboard — wired into the Business Workspace nav and org-creation flow, confirmed rendering live
51 new automated tests, including the two load-bearing properties written before their implementation: citizen-record access requires an actual assignment, and a published policy knowledge base entry can never be mutated in place. Nothing on inaya-mobile yet. No government certification, accreditation, FedRAMP authorization, or jurisdiction-specific compliance claim exists or is claimed anywhere.	
INAYA NETWORK	INAYA-DEV-2026-V1

SECTION 21

Institutional Trust Infrastructure Reference (September 2026)

FUNCTION / ROUTE	PURPOSE
evidence.js: getEvidenceTrail, verifyOrgEvidenceIntegrity, exportEvidencePackage	Read-only merge of listOrgActivityForRecord() + linked aiActionRequests into one timeline; integrity check is a pass-through to auditChain.js's verifyChainIntegrity()
org-trust.js: proposeTrustRelationship, respond, revokeTrustRelationship, isTrustedAccess	New orgTrustRelationships collection; accept/reject filtered by findOneAndUpdate({toOrgId: caller}) so an org can never approve its own proposal; isTrustedAccess is a query primitive no other module calls yet
POST/GET /api/orgs/trust-relationships, PATCH /api/orgs/trust-relationships/[relationshipId]	Session-authenticated cross-org trust CRUD
api-keys.js: generateToken/hashToken convention, requireApiKey	Raw key shown once at creation, only hash persisted; resolves to a synthetic owner-level membership permanently bound to the key's own orgId — no request parameter can override it
POST/GET /api/orgs/api-keys, DELETE .../[apiKeyId]	Session-authenticated key management, owner/admin only
GET /api/public/v1/audit/verify, /api/public/v1/evidence, /api/public/v1/permissions/check	API-key-authenticated (Bearer). evidence requires both recordType+recordId (no org-wide dump); permissions/check reports an org-level capability flag, not per-user RBAC
institutional-attention.js: getAttentionItems	AI-tool-only (get_attention_items in ai-os-router.js), not a REST route; merges pending-approvals (re-checked against the real approval gate), overdue tasks, pending documents
src/app/trust/page.js — /trust	Public, unauthenticated. Client-side verifier reimplements auditChain.js's hash formula via Web Crypto; auto-detects genesis vs. scoped exports (fixed in 9345df5) and reports the correct guarantee strength for each

Zero mainnet references across every file/route touched by this section, confirmed by direct grep. Every module ships a co-located test file (org-trust, evidence, institutional-attention, api-keys), none mocked against a fake database.

SECTION 22

Autonomous Resilience Layer Reference (September 2026)

FUNCTION / ROUTE	PURPOSE
resilience-policy.js: createPolicy, updatePolicy, evaluateRtoRpo	requiredRTOMinutes/RPOMinutes + critical-asset categories + test frequency; evaluateRtoRpo is a plain actual<=required comparison, no fuzzy scoring
resilience-canary.js: ensureResilienceHome, createCanaryAsset	Synthetic org-owned documents (isResilienceCanary:true, hidden from normal browsing) pushed through the real InayaKernel.disperseAndSlice() + real pinning-provider .pin() + real backupEngine.replicateShard() — never a mock of these functions
resilience-orchestrator.js: runResilienceTest, runScheduledResilienceTests	13-step test: real reconstruction via InayaKernel.reconstructAndDecrypt(), real getPinStatus() dependency check, real getDocumentAccessLevel() permission check, wall-clock RTO, replica-staleness-based RPO; atomic RUNNING-status duplicate-run guard
GET /api/cron/resilience-tests	CRON_SECRET Bearer-gated daily sweep, registered in vercel.json; per-policy try/catch so one broken policy never blocks the rest
resilience-status.js: deriveState	Pure function → Unknown / Test Due / Verified / Failed / Degraded, unit-tested directly; a PAUSED policy always reports Unknown regardless of test history
GET /api/orgs/resilience/status, POST /api/orgs/resilience/policies/[policyId]/run	Dashboard read + manual 'Run now' (canManageOrg-gated, triggeredBy:'manual' vs. cron's 'schedule')
ai-resilience-tools.js: get_resilience_status, get_last_test_result, list_failed_assets, explain_test_failure, trigger_resilience_test_now	First four are structurally read-only (no mutation code path exists); trigger_resilience_test_now can only start the same non-destructive test the cron runs, kept outside Guarded Execution because there is no destructive-workflow surface for it to gate

Two small, additive fixes to the pre-existing backupEngine.js's replicateShard() (expose providerRef; accept an explicit primaryProviderRef) were required and made in this same pass, both non-breaking. Verified live against real Filebase network calls during development.

SECTION 23

Node Operator Dashboard Reference (September 2026)

FUNCTION / ROUTE	PURPOSE
nodeOperatorAuth.js: loginWithWalletSignature, requireNodeSession, linkWalletToSession	New node_sessions collection, 12h cookie (inaya_node_session), reuses the existing verifyNodeAuth() signature check unchanged; fleet-linking requires the linked wallet's own signature
nodeUptimeHistory.js: getUptimeForWindow	Hourly-cron-populated node_uptime_snapshots; sets insufficientData:true and reports the real coverage window rather than backfilling or estimating history it doesn't have
nodeChainReads.js: getOnChainNodeInfo	Real ethers.JsonRpcProvider reads against the deployed node-registry contract (same address the existing settlement-release cron uses); tier/commission explicitly sourced on_chain since the off-chain Mongo tier field is set once and never revisited; fails honest-empty, never fabricates a tier
GET /api/nodes/operator/{me,uptime,qualification,rewards,events,network,fleet}, POST .../link, .../login, .../logout	network is the only unauthenticated route (public aggregate stats); every other route scopes strictly to the session's own wallet — no client-supplied node ID is ever trusted
GET /api/cron/nodes-snapshot	Bearer CRON_SECRET-gated hourly snapshot job, registered in vercel.json
src/app/operator/page.js + OperatorOverview/Uptime/Qualification/Rewards/Events/Network/Fleet.js	Seven-tab dashboard, standalone MetaMask connect-and-sign login independent of the dApp's WalletContext and Business Workspace's session

No claim/release control anywhere in the dashboard — settlement stays fully automated by the pre-existing timelock cron. Public /network reports off-chain tier distribution, not a live per-node chain read, an explicitly disclosed scalability trade-off. Tested with real ethers.Wallet signatures, treated internally as the highest-risk new surface in this pass.

SECTION 24

Four New Business Primitives Reference: Sign, DePIN Storage, Escrow, Attestation (September 2026)

FUNCTION / ROUTE	PURPOSE
signing-workflow.js: recordSignature, verifyInayaSignSignature, anchorSigningRequestOnChain, verifySigningRequest	Wallet path = real ethers.verifyMessage() ECDSA check, 5-min freshness window; session_consent path is explicitly labeled, never presented as equivalent; anchor is a real, optional transaction (batchRegisterAssets on the existing custody contract) over a completion-certificate hash, not a per-signature on-chain record
storage-manager.js: registerOrgStorageNode, getOrgStorageOverview, reportOrgStorageNodeHealth	New org-scoped control plane, confirmed unconnected to the public node-operator registry; eligibility is a static 'routing_not_yet_supported'; allocation/replica figures reused as-is from orgPlans.js + backupEngine.js, not recomputed
escrow-workflow.js: transitionEscrow, confirmMilestone, releaseMilestonePayment	State machine only flips a status field; the ONLY function moving money is releaseMilestonePayment, reachable only after a proposeAiAction() Guarded Execution approval plus the standard 36h SETTLEMENT_DELAY_MS; writes a real payments row, re-validates dispute status at execution time
financial-attestation.js: buildCanonicalDataset, generateAttestation, verifyAttestation	sha256 hash-commitment over real permission-scoped, currency-converted financial records; only revenue_threshold/expense_threshold implemented, solvency_threshold explicitly rejected (no assets/liabilities data model exists); verification independently re-runs the query and recomputes the commitment
POST/GET routes under /api/orgs/sign, /api/orgs/escrow, /api/orgs/storage, /api/orgs/attestations (~19 routes)	SignInView, EscrowView, StorageManagerView, AttestationsView in src/components/business/ — real nav entries in page.js's Operations group, real org-scoped API-backed screens

Each module's own header comment states plainly what's real vs. not: Escrow is never described as on-chain or non-custodial anywhere in its code, routes, or UI; Attestation names the specific ZK libraries confirmed absent from the full dependency tree to avoid that label being misapplied; DePIN Storage Manager's region preferences are always labeled Declared, never Active. A real nested-object canonicalization bug found while testing Attestation was also fixed in Sign's anchor certificate, which shared the same flaw. Test coverage: inaya-sign, milestone-escrow, depin-storage-manager, financial-attestation.

SECTION 25

Business Workspace Operational Enhancements & Guided AI Assistant Reference (September 2026)

FUNCTION / ROUTE	PURPOSE
finance/invoices/[invoiceId]/pdf/route.js	Returns print-styled HTML, not a binary PDF — the route's own comment confirms no PDF library exists in the app; values pulled verbatim from the invoice record, never recalculated
currency.js: convert, SUPPORTED_CURRENCIES, USD_RATES	Static, explicitly dated reference-rate table (RATES_AS_OF), not a live FX feed by design; returns an error rather than guessing on unsupported pairs
inventory.js: transferStock, recordStockMovement	Two ledger movements + a materialized stock_levels update sharing one transferId; a real compensating ADJUSTMENT reversal if the second leg fails after the first commits
purchase-order-workflow.js: updatePurchaseOrderItems	Atomic findOneAndUpdate({status:'DRAFT'}) guard — editing outside DRAFT returns a real 409; reruns the same validator the create route uses
procurement/orders/[orderId]/email, procurement/reports	Real supplier email with real line items via the existing sendEmail() primitive (documented no-op if RESEND_API_KEY unset); 5 real report types incl. CSV export
insights route.js + business-insights.js: computeBusinessInsights	Real custom start/end date range on top of preset windows, validated (endDate<startDate → 400), computed via the same permission-scoped getAccessibleScope() every other route uses
business-chat/route.js — Gemini (gemini-3.5-flash-lite) via @google/genai, Groq fallback	Real, production-hardened LLM call: per-call timeout, request-level safety budget, retry-on-503 — added after an observed Gemini-queuing production issue
guided-workflow-catalog.js (static data, no LLM calls) + guided-tasks.js (state machine)	~10 hardcoded workflows; the model only ever relays pre-authored instruction text verbatim — structurally prevented from inventing step text; the task's own bookkeeping collection is the only thing it ever writes to
GuidedTaskPanel.js + inaya:guided-nav / inaya:guided-po-created custom events	Step advancement fires from real navigation events, real UI success events, or a manual 'I did this' confirmation — never the model's own assertion that a step is done

Creating a new business record (contact, deal, PO, document) has no automated tool path at all — the assistant can only guide a human through the real UI. Any actual mutation it can propose still routes through the pre-existing Guarded Execution approval-plus-36-hour-delay pipeline described in §26/§31. No new mutation surface was added by the guided-task feature itself.

SECTION 26

Multi-Cloud Storage Compatibility & Enterprise Storage Governance Reference (September 2026)

FILE / ROUTE	PURPOSE
s3-compatible/sigv4.js: verifySigV4Request, parseAuthorizationHeader	Real AWS SigV4 verification — canonical request → string-to-sign → derived signing key (HMAC chain) → signature; 15-min clock-skew window; UNSIGNED-PAYLOAD support
s3-compatible/azureAuth.js: verifySharedKeyRequest, s3-compatible/azureAuthMiddleware.js: authenticateAzureRequest	Azure Shared Key verification + real Microsoft Entra ID Bearer-token path (live Graph verification via integrationProviders/microsoft.js, mapped to org_members by email)
s3-compatible/credentials.js: issueS3Credential, checkScope, normalizeScope	Credential issuance with optional scope {bucket, prefix, operations, expiresAt}; checkScope() is the single enforcement function, called centrally in auth.js/azureAuthMiddleware.js after signature verification
s3-compatible/store.js, walletStore.js: putS3Object, deleteS3Object, assertNotProtected	Org- and wallet-side translation layers over org_documents/metadata_files; assertNotProtected() is the one chokepoint both the overwrite path and the delete path call through for Object Lock/Legal Hold enforcement
s3-compatible/store.js: putBucketVersioning, listObjectVersions, restoreObjectVersion	Bucket-level versioningStatus (Unversioned→Enabled→Suspended, one-way); restore copies an old version's bytes forward as a new current version rather than resurrecting the old version id
s3-compatible/store.js: putObjectRetention, putObjectLegalHold, putLifecyclePolicy, runLifecycleEnforcement	Object Lock (requires Versioning Enabled first; retention can extend, never shorten), Legal Hold, and per-bucket lifecycle rules with real enforcement that skips locked/held objects
api/s3/**/route.js, api/azure/**/route.js	Full S3 and Azure REST surfaces, including real sub-resources (?versioning, ?object-lock, ?versions, ?legal-hold, ?retention, ?versionId=, Put Block/Put Block List)
api/cron/s3-lifecycle/route.js	CRON_SECRET-gated route calling runLifecycleEnforcement() — same auth convention as api/cron/nodes-snapshot; registered in vercel.json's crons array
api/orgs/s3-compatible/manage/route.js, S3CompatView.js	Session-authenticated Business Workspace management API + console — separate from the SigV4-signed protocol surface, so Business Workspace never needs to hold/sign with an S3 credential
backupEngine.js: replicateShard (now called from putS3Object)	Closes a real, previously-existing gap: S3/Azure-compatible objects were pinned but never registered with the existing check-pins/verify-integrity/recovery crons
inaya-drive-helper/src/main.rs, sigv4.rs, s3client.rs	Standalone Rust crate (GPL-3.0, isolated on purpose) implementing WinFSP's FileSystemContext trait over a Rust port of the SigV4 signer, backed by the real /api/s3 endpoint
inaya-desktop/src-tauri/src/lib.rs: mount_inaya_drive, unmount_inaya_drive	Tauri commands that only spawn/kill inaya-drive-helper.exe as a child process — no GPL dependency in inaya-desktop's own Cargo.toml

Test coverage: test/s3-compatible-capabilities.test.mjs (19 tests: scope enforcement, versioning, Object Lock, Legal Hold, lifecycle, health registration) plus the pre-existing test/s3-compatible-sigv4.test.mjs (9) and test/s3-compatible-store.test.mjs (9) — all passing, zero regressions. Full writeup: docs/storj-inspired-storage-capability-expansion-report.md and docs/multi-cloud-storage-compatibility-report.md.

SECTION 27

Storage Interoperability Expansion Reference — Empty Folders, Migration Tooling, Signed URLs & GCS Auth (September 2026)

FILE / ROUTE	PURPOSE
inaya-drive-core/src/lib.rs, s3client.rs, sigv4.rs	New standalone Rust crate extracted out of inaya-drive-helper: the SigV4-signed S3 client and signer, with zero Windows/WinFSP dependency, now linked unmodified by both inaya-drive-helper (Windows/WinFSP, GPL-3.0) and the new inaya-drive-helper-linux (Linux/FUSE via the MIT-licensed fuser crate, UNLICENSED) — one real client implementation, not a per-platform rewrite. Each platform's own filesystem-trait binding (WinFSP's vs. FUSE's) stays separate, since the two OS filesystem APIs aren't interchangeable
s3-compatible/store.js: createS3Folder, getS3FolderInfo, deleteS3Folder, renameS3Folder	A real, durable folder record in a new s3_folders collection (mirroring the pre-existing Business Workspace metadata_folders shape) — a folder with zero objects is now listable via ?prefix&delimiter, where previously only shared object-key prefixes existed and an empty folder was invisible
inaya-migration-agent/bin/inaya-migrate.mjs	Standalone Node CLI: --source aws azure gcs, resumable via a local --manifest file (skips already-verified keys on rerun), --dry-run, --objects for a scoped subset; every source/destination credential stays local to the invoking machine, never transmitted to or stored by Inaya
s3-compatible/signedUrl.js: createSignedUrl, verifySignedUrl	Query-string presigned GET URLs (X-Amz-Expires/X-Amz-Signature convention): reuses the existing SigV4 signing key derivation rather than a new secret, rejects on expiry, method mismatch, or any tampered query parameter
googleAuth.js: verifyGoogleIdToken, s3-compatible/auth.js (Google ID-token branch)	Google Sign-In accepted as a direct auth path on the S3-compatible endpoint: verifies the Google ID token's signature and audience server-side, then resolves it to an existing org_members row by verified email — no separate Google-only permission tier or credential type
middleware.js: bucket-from-hostname rewrite, gated by S3_COMPAT_VIRTUAL_HOST_BASE	Virtual-hosted-style addressing (bucket-name.base-host/key, matching real AWS/GCS convention) — the Host header is parsed and rewritten to the existing path-style route before any other S3/Azure code runs, so the protocol surface itself is unchanged; unset (off) by default
evidenceExporter.js: buildEvidencePackage, canonicalizeForExport	Reads an org's own existing audit chain and storage-protection settings (Object Lock, versioning, retention) into one canonicalized, hash-stamped package — no new data is generated or claimed, only exported
evidencePdf.js: renderEvidencePdf	Deterministic PDF rendering of the same package for a non-technical reader, hash printed on the document itself so a paper copy can still be checked against the JSON export

Test coverage: test/s3-compatible-empty-folder.test.mjs (15 tests: placeholder-key creation/listing/deletion, cross-platform key-prefix handling, mount-restart persistence), test/signed-url.test.mjs (9: expiry, tamper rejection, method restriction, real wall-clock expiration — not a simulated clock), test/evidence-exporter.test.mjs (5), test/migration.test.mjs (6) — all passing against the real dev database, zero regressions in the pre-existing S3/Azure-compatible suites.

Two corrected findings, kept honest rather than restated as always-true: Google's own "gcloud storage" CLI was earlier reported unable to reach a third-party endpoint like Inaya's — it can, through an official (Google-labeled "unstable") configuration option, verified with real uploads and downloads. The older "gsutil" tool still cannot, due to a genuine bug in Google's own software; a workaround narrowed it further but did not fully resolve it. Remaining disclosed limitation: macOS Drive support is written but not yet compiled or tested on real Mac hardware. Full writeup: docs/inaya-drive-empty-folder-creation-report.md, docs/enterprise-adoption-and-market-reach-report.md, and docs/gcs-compatibility-extension-report.md.

